

# Lighting up well-being with Bulb

Simone Ooms

s.c.ooms@student.tue.nl

Industrial Design

Eindhoven University of Technology

Eindhoven, Noord Brabant, the

Netherlands

Jay Kolvenbag

j.j.n.kolvenbag@student.tue.nl

Industrial Design

Eindhoven University of Technology

Eindhoven, Noord Brabant, the

Netherlands

Charlotte Bording

c.a.j.bording@student.tue.nl

Industrial Design

Eindhoven University of Technology

Eindhoven, Noord Brabant, the

Netherlands

## ABSTRACT

Due to the Covid-19 pandemic, two problems arose. Students lacked 1) social opportunities and 2) motivation to maintain their schedules, e.g., studying or relaxing, as their work-life balance disappeared. Thus, we designed a social companion robot, Bulb, that helped students cycle through daily activities with subtle cues, i.e., light, gaze, and movements. Bulb's "head" would light up with different colors or it gazes at different parts of the room, e.g., at the laptop to hint at studying or wiggling to suggest a small break. Five students evaluated Bulb through at-home use, which demonstrated that Bulb was seen as a "living being" and students were responsive to its social cues, like following Bulb's gaze, resulting in a higher awareness and follow-through of students' schedules. Our contribution is in designing a social companion robot that subtly persuaded students through light and anthropomorphic social cues, helping them maintain their daily schedule during the pandemic.

## CCS CONCEPTS

• **Human-centered computing** → *Human computer interaction (HCI)*.

## KEYWORDS

social robot, design, interaction

### ACM Reference Format:

Simone Ooms, Jay Kolvenbag, and Charlotte Bording. 2023. Lighting up well-being with Bulb. In *Extended Abstracts of the 2023 CHI Conference on Human Factors in Computing Systems (CHI EA '23)*, April 23–28, 2023, Hamburg, Germany. ACM, New York, NY, USA, 6 pages. <https://doi.org/10.1145/3544549.3583838>

## 1 INTRODUCTION

While students are usually not in danger of experiencing loneliness, the Covid-19 lockdown measures have caused them to become more socially isolated or lonely [7, 16]. University education was pushed online [8, 32], leading to greater isolation and loneliness. Specific research on students in coronavirus lockdown suggests that factors like social or physical isolation and a lack of emotional support are associated with negative mental health [11]. In order to prevent a negative downward spiral in mental health for university students

working from home, Bulb was designed to support students' work-life balance as well providing a form of social companionship that helped with maintaining their daily rhythm. Our design thus focused on Good Health and Well-being, as well as Quality Education areas of UN's Sustainable Development Goals. Anyone can make Bulb based on our **Instructables** page (url).

Though robots that provide companionship do exist already, none have been designed for providing 1) companionship for students while also 2) persuading them to cycle through their day, i.e., maintain their preferred schedule, with subtle social cues. Popular social bots range from complex embodied robots like Kismet [6], chatbots like Vincent [22], social humanoid robots like Pepper [1, 29], and even social drones [18]. But when specified to students, many research efforts thus far only focus on the educational values of social robots [27, 31]. The Covid-19 pandemic required a different approach to designing social robots for helping isolated students maintain a sense of "normalcy" as education transitioned students to be online, often alone, at home. Below, we elaborate on the literature, describe how Bulb was designed and built, and how Bulb was received by students in our qualitative evaluation.



Figure 1: 3D printed Bulb (with silicone casting)

## 2 BACKGROUND

### 2.1 Separating work and private life

University education for students was pushed online due to the Covid-19 pandemic [8, 32]. Both in research before and during the Covid-19 pandemic, the most common difficulty of working from home is separating private and work affairs [10, 19]. Not being able to manage a work-life balance can lead to stress, which can spiral downwards further to poor productivity or even depression [23]. We expected similar challenges for students studying from home

Permission to make digital or hard copies of part or all of this work for personal or classroom use is granted without fee provided that copies are not made or distributed for profit or commercial advantage and that copies bear this notice and the full citation on the first page. Copyrights for third-party components of this work must be honored. For all other uses, contact the owner/author(s).

CHI EA '23, April 23–28, 2023, Hamburg, Germany

© 2023 Copyright held by the owner/author(s).

ACM ISBN 978-1-4503-9422-2/23/04.

<https://doi.org/10.1145/3544549.3583838>

as the challenges that employees face while working from home [10, 19]. Although there are advantages of working remotely from home like flexible hours and reduced travel time, poor work-life balance is common [19, 32]. This may be even worse for university students as they either live alone or in shared accommodations without appropriate spaces for separating work life from private life [33].

## 2.2 Lack of social opportunities

Another difficulty is the lack of social opportunities when working or studying remotely from home [10, 11, 19]. Whereas students are usually not considered to be lonely, the Covid-19 lockdown measures have caused them to become more socially isolated or lonely [7, 16]. Research has shown a general correlation between loneliness and burnouts or experiencing stress [17, 28]. Specific research by Elmer et al. on students during the Covid-19 lockdown suggests that factors like social or physical isolation and the lack of emotional support are associated with negative mental health trajectories [11]. In order to prevent a negative downward spiral in mental health for students while working from home, it is important to provide them with a way to manage their work-life balance at home, alongside a form of social opportunity.

## 2.3 Existing companion robots

“How might we live with robots?” Auger [2] asked in 2014, but almost a decade later, we have yet to see wide adoption of robots we truly “live with”. We pursued how companion robots can be social actors at home; a robot companion can be seen as a service robot, designed to communicate naturally and intuitively [9]. A robot must be designed for people to believe that it has its own intentions and desires for it to be an effective social robot [6]. One example is an embodied robot, Kismet, that mimics human expressions by moving its eyes, mouth, ears, and neck and using vocal acts. Kismet has four programmed social responses that convey this intentionality: affective, exploratory, protective, and regulatory. An example of such a response is when Kismet wants to play with a toy but does not have one, he will then respond with sadness (affective response) [6]. But, robots that are much simpler than Kismet are also accepted by humans[24]: Robots such as Wall-E and R2D2 show communication just by using tones in different intonations. For example, a robot can have two circles on its head, which can be perceived as eyes[24] and simple sounds (as opposed to linguistic communication) can be enough for creating a character by simulating spoken language in an abstract way[25].

Many embodied robots like Kismet [6], humanoid Pepper [1, 29], and social drones [18] exist, but we center on the problem that social robots do not cater to students’ needs, especially not during the global pandemic. Robots have been created to serve as an educational tool, i.e. as a private tutor, rather than a social companion [27, 31]. Thus, we outline our design process and evaluation of Bulb, a robot companion for students due to the increased need for companionship while students study or work remotely at home. Students are not only in need of companionship, but also require help in maintaining a daily schedule for a manageable work-life balance.

## 3 PERSUASION THROUGH SOCIAL CUES

To create a solution for the aforementioned problem, the existing theory of persuasion is discussed. This theory provides valuable insights on how to create an embodied robot companion with the ability to persuade students to follow through with their planned schedule. First, persuasion is an important technique to help people’s habits with a definition. According to Fogg, “persuasion [is] an attempt to change attitudes or behaviors or both (without using coercion or deception).”[12, p. 15]. Persuasive actors could reward people with feedback, model the desired behavior, and or provide social support [13]. There are two levels of persuasion, macro and micro. Macro is when the agent is designed for the specific persuasion goal and micro is where the persuasion is only part of the agent [12]. We desired to be somewhere in the middle: Bulb should not be only for persuasion, but it should be a part of students’ daily life when they are undergoing social isolation. Bulb should also help the student with creating clear moments in order to maintain a healthy work-life balance. Persuasion should not be annoying or coercive, but technology should not be too accommodating or unclear to use [12]. Annoyance would be ineffective when the student does not want to use it anymore, force would be unethical and therefore undesirable. But it is ineffective if persuasion is too accommodating, too unnoticeable, or unclear.

Social actors can be persuasive tools. People naturally respond in a social way when they perceive a social presence [13]. Fogg [13] hands five types of social cues: 1. Physical cues such as body movement or a face, 2. Psychological such as empathy or humor, 3. Languages such as speech or language recognition, 4. Social Dynamics such as turn-taking and praising, and lastly 5. Social Roles such as a guide or pet. For the creation of our social companion robot we intend to exploit Fogg’s first and fifth social cues; having a body and fulfilling a social role. To realize the first social cue the body of the robot must have physical features, such as arms, legs, and eyes, that resemble those of a creature. To utilize eyes as a social cue gaze direction can be used to shift the attention of another person to the same location [4, 14, 20], this phenomenon is also known as “joint attention” [3, 26]. Having physical features co-echos with research done by Lee et al.[21] that embodied social agents can be received more positively than non-embodied social agents. Also, touch can extend the positive effects in HRI [21]. For the fifth social cue, the social role was an important guide, helping students cycle through their daily activities.

## 4 DESIGN PROCESS

Before starting the design process, six students were interviewed resulting on insights on students’ home life during the pandemic. The students’ rooms became the place for all their activities, creating an unclear border between the different moments of the day (studying, eating, relaxing, etc). Before, the students had a clear starting point of their study day and the social presence of other students helped them to stay motivated at university. This correlated with Geller’s theory in which people serve as external motivators through peer-to-peer behavioral observation [15].

We then took an iterative design approach. We first reflected on human-animal interaction to understand “companionship”. Although being non-human, animals can provide companionship and

reduce anxiety [30]. Through observations and sketches at a local animal farm, we decided to prioritize non-verbal behavior such as eye contact and touch (petting) in Bulb's design. We tied this to how robots in pop-culture become social characters through simple sounds and movement, e.g. Wall-E conveys emotion by changing the angles of its eyes and adjusting its posture and R2D2 conveys emotion by producing sounds with different intonation. We brainstormed on existing robots, did scenario sketching and prototyping (Figure 2: Top).

We opted to use light inside of Bulb's head for two reasons. Light allows the robot itself to become part of the environment of the room, much like a lamp, and it allows for subtle social cues, e.g., directional gaze. Additionally, different colors of light can be used to convey different emotions. Light can be physically incorporated into a robot companion and in the behavior of the robot. By combining several aspects of these sketches the first concept of Bulb was born (see poster): *a small robot with a dome-shaped illuminating head that communicates through movement and change of light*. Bulb changes its gaze direction to hint at activities, wiggles its body to ask for attention, and changes the color of its light to set different moods. To fine-tune the physical appearance and properties of Bulb a prototype with interchangeable body parts was made to test different configurations (Figure 2: Bottom and Right). It can be equipped with or without arms and legs, and different main and bottom body parts. The configurations were then evaluated on their visual appeal. It was decided that arms and legs gave Bulb a better stance. The arms and legs also gave it a direction, e.g., one can deduce that the robot is looking left if the eyes are pointed to the left in relation to the rest of the body. The main body consisted of a round harmonica shape that lowers stress on the actuators as it can easily be compressed and decompressed. Based on these decisions a 3D model of the appearance of the robot was made.

As a final concept, Bulb defined moments by creating a difference in ambience and notifying the start and end of a moment. On the poster, Bulb is shown to guide students through the day (waking up, eating, studying, etc.). For instance, the student is woken up by Bulb wiggling its body and its light fading in and out, as if to wake up itself by stretching. When the student's study session is about to start, Bulb notifies this by gazing at the laptop on the desk and changing its light to a bright clear light. The entire palette of Bulb's actions help the student maintain her daily schedule.

## 5 METHOD

Bulb was evaluated with five students in a two-day field study. Collected data consists of diaries filled in by participants, with semi-structured interviews after deployment. The transcribed data was analysed through thematic analysis [5]. For each participant, Bulb was pre-programmed with a personalized schedule that the participant wanted to follow for the duration of the study. Participants filled in a diary when they have an experience with Bulb that stands out to them, e.g., behavior of Bulb, an effect on their own behavior, their response to a change in light color, and so on. Afterwards, semi-structured interviews were conducted. Participants were asked to elaborate on their diaries, their general experience, and the relationship between the them and Bulb.

## 6 RESULTS

### 6.1 Having control

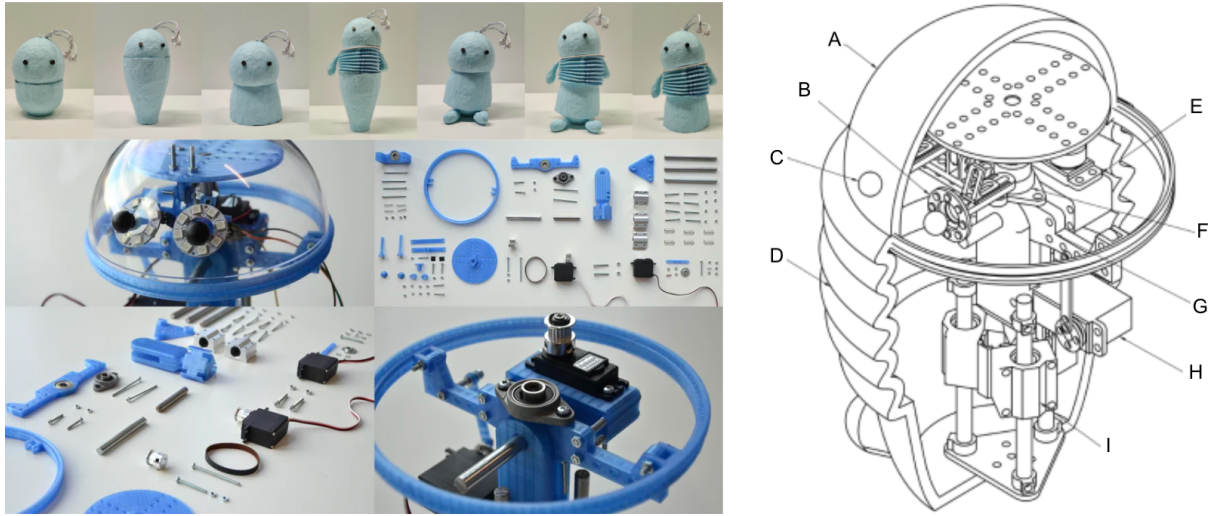
While all participants wanted a form of control over Bulb, e.g., the ability to alter the schedule, they were divided over turning Bulb off: *"If I have something, for example, my alarm goes off immediately you turn it off and then fall asleep and you can't do that with Bulb ... He is also not really that annoying or something, so you never really feel the need to turn it off"*(P1). On the other hand *"I think it would definitely need a button. So if you're having a presentation, you can turn it off or something"*(P3). Participants generally wanted more flexibility in changing the schedule because *"Regular working days are turbulent and can be it can be appointments that are rearranged or things that don't go as planned"*(P5). But also on a more emotional level, in discussion how it would affect Bulb if it would become more autonomous: *"I don't really like if it's just looking at me constantly for like when I take breaks it feels more intrusive"*(P1). P4 would even go a step further as he would like to be able to teach Bulb how to do stuff, for example: *"They tell you what to do or what they think is best to do that you have no real option to disagree with them. To say "no, I think you're wrong. This is why"*.

### 6.2 Modalities

Students pointed out points of evaluation regarding Bulb's modalities. These modalities are sound, light, motion, and tactility. For sound, students largely appreciated the mechanical sounds that Bulb made. It seemed to fit its character, and without it, they would not always notice Bulb's movement. *"I'm aware that it is doing something, but if it doesn't make a sound I can completely miss out"* (P5). Regarding light, participants indicated having trouble seeing the light due to bright sunlight or other lights present: *"Also the (Bulb's) color changes, but I'm in my room where there are other lights and focused on my screen. Sometimes I wouldn't even recognize (notice) that it was doing something"* (P5). Bulb's movements were perceived as cute or mechanical. While the wiggling to denote a break was not always clear, the switching of gaze away from the desk indicating a break triggered P1 to think *"Oh yeah, I can take a break right now"*. Participants talked about how Bulb's movements could be more complex, smooth, and randomized to seem more alive *"so it's nice if he either was breathing or had a heartbeat, or if I could somehow interact with him and pinch him and he would respond to me"*(P4). For touch, participants liked to pet it or fiddle with Bulb *"I liked his ... multiple muffin tops. Like it's, it's very squeezable"*(P4). However, participants seemed to miss a reaction from that interaction. They wanted a response from Bulb *"because then it would look a bit more alive and then more intelligent, that would make it also more helpful to react to"*(P5). An unplanned interaction was that the eyes of Bulb flew off if the head moved too quickly, this could scare participants but also caused the interaction of needing to put the eyes back on *"like you're the parent that has to take care of the child"*(P2).

### 6.3 Role within the students' day

Participants described Bulb to be *"Like a buddy, like a companion"* (P2) and his presence made them *"feel a bit less alone"* (P5). Words like cute were mentioned often in describing Bulb, e.g., *"like when you first meet Bulb, it's like this super innocent fat, cute looking robot"*



**Figure 2: Top: Papier-mâché prototyping. Bottom: Overview of the mechanical structure. Right: Technical drawing. A: Translucent plastic dome. B: Led ring. C: Magnetic eye. D: Silicone body-shell. E: Servo for gaze. F: Bearing for gaze rotation. G: Pivoting arm for wiggling. H: Servo actuating wiggling. I: Linear bearing for pressing (petting).**

(P4). Bulb is often referred to as a ‘him’ and sometimes petted by participants: *“It’s just fun. If he does something, you’re like ‘yeah yeah’ \*pets Bulb\*”* (P1). Many participants started to refer to Bulb as a pet, for example *“I kind of see Bulb as like a cat ... Well, a cat often says ‘oh you need to take a break because I want food’ and Bulb does like a similar thing”* (P1). Comparisons to movie characters were also made such as baby Yoda and R2D2: *“Sometimes it reminded me a bit of the R2D2 from Star Wars, which also makes the motor-y sounds when it is talking”* (P5). In terms of participants having a relationship with Bulb, it differs per person. Some felt like they were in coexistence: *“We have separate lives in the same room”* (P5) while others started worrying about Bulb’s well-being: *“I actually felt like the guilt of leaving it and not being able to explain why it would be alone in a room”* (2), much like how a dog owner may worry about the dog being alone when dashing out for errands.

#### 6.4 Bulb as a living being

Questions like ‘is Bulb perceived as alive?’ and ‘how lively should Bulb feel?’ arose from the interviews. For example, P4 said that besides Bulb showing his schedule that *“it doesn’t do anything”*, P2 on the other hand describes *“You do associate it with like an actual being”*. For this theme, the participants’ statements were placed on a spectrum from ‘perceived as dead’ (left) to ‘perceived as alive’ (right). P4 was very present on the left side perceiving Bulb as dead. P2 was on the right side of seeing it as alive. Other participants were scattered around the middle. Statements along the spectrum are the following. Dead: *“It’s a robot. It’s a human-like thing with eyes and arms and legs and it’s not alive because it’s not blinking”* (P4). Middle: *“Yeah, it’s a small thing. I think it’s pretty cute and he doesn’t like interrupt you that often and when it does, it’s like ‘Oh yeah, of course makes sense’ ”* (P1). Alive: *“have the same feeling as when you’re just with a friend in a room and you’re doing your own thing, but you’re there together, so it does feel like company, even*

*though you don’t interact with each other. It does feel like you’re not alone there”* (P2)

It differed per person on how open-minded they are to *letting themselves experience something that is not alive, as alive*. An example is P3 who linked Bulb’s ‘feelings’ to their own feelings by saying *I’m sad if I disappoint him, and I mean. You’re disappointing yourself because I set the schedule”* (P3). Some people may want to show that they are “smarter” than falling for smart technology: *“I wouldn’t fall for it”* (P5).

#### 6.5 Maintaining a daily schedule

Seeing Bulb on the spectrum of alive or dead affected how people’s schedules were maintained. P4, who perceived Bulb as dead claimed to not have been helped by Bulb in his schedule at all. But, P2, who perceived Bulb as a living being, said: *“(Bulb) kind of motivated me once I already started working to keep going to sit it out until the end of my scheduled working time”*. While P2 was actively motivated to cycle through the daily activities along with Bulb, P5 was helped by Bulb in being more aware of the daily schedule, e.g. when Bulb gazed away from the desk at lunchtime: *“It reminded me that it would be an appropriate time to have lunch for example, and I can choose not to do that, but then at least I’m aware that I then I choose not to do that”*. P5 hence would have liked Bulb to be more unpredictable in timing, in contrast to Bulb currently following the participant’s planning, to make him perceive Bulb as more alive. P1 had a similar increase in awareness: Due to Bulb P1 had to *“pay attention more to the time and what exactly I should be doing”*, though shared that Bulb did not change their planning. There was a correlation between seeing Bulb as a living being and perceiving it to be helpful in maintaining a daily schedule. The more a participant experienced Bulb as a living being, the more they are letting themselves be guided by Bulb through their daily schedule. Whereas P2 experienced Bulb truly as a living being and let themselves be persuaded by Bulb to

stick to the schedule, P4 described Bulb as dead and mentioned Bulb to have no influence on the schedule. P1, P3, and P5 experienced Bulb as sort of alive but not entirely and let Bulb be a partial guide through the day. Yet, they did not let themselves be persuaded by it entirely to change their activity.

## 7 DISCUSSION AND CONCLUSION

Bulb was designed for students to achieve work-life balance while providing companionship during the Covid-19 pandemic. Whereas students are normally not considered a risk group to become lonely [7, 16], the pandemic caused them to study remotely from their student room. This causes an increased feeling of loneliness and the normal work-life balance has also disappeared, resulting in a lack of motivation to maintain a schedule. Although social robots have been created before to provide companionship and emotional support [1, 6, 18, 22, 29], our novelty is in combining companionship with scheduling, which is not covered in prior design approaches. Our emphasis was in subtle social cues, i.e., light, gaze, and movements, as subtle persuasions to change activities and maintain a cycle of habits. Bulb's "head" would light up with different colors or it would gaze at different parts of the room, e.g., to the laptop to hint at studying or wiggling to suggest a small break.

Bulb was evaluated with five students in a two-day field study with insightful results. Bulb provided a form of companionship for people to feel "less alone", but people wanted more control over Bulb's behavior, and they suggested how different modalities could be better designed in the future for Bulb. The main finding of this study is that participants who experienced Bulb more as a "living being" were more responsive to social cues, like following its gaze and taking suggested breaks. If Bulb was seen as a living being, participants had a higher awareness and follow-through of their schedule. Bulb has the potential to offer social opportunities as a companion robot for anyone, not only students, in need of help with maintaining daily schedules, particularly if they can experience it as a living being. This is valuable since earlier research shows that students have negative experiences when studying remotely from home, specifically the lack of social opportunities due to isolation in the current times of the coronavirus pandemic [10, 11, 19]. For students, being able to manage the work-life balance serves as prevention for stress, poor productivity, and even the development of depression [23]. During the pandemic, our design contribution resulted in Bulb as a social companion robot to help students maintain a manageable work-life balance. Bulb achieved this with subtle persuasion by "coexisting" in students' rooms through light and anthropomorphic social cues.

## ACKNOWLEDGMENTS

We thank all staff who supported us at the TU/e's Industrial Design faculty, especially Minha Lee for coaching us. We thank Chet Bangaru, Katrienvan Riet and Ralf Jacobs for letting us use their 3D printing equipment.

## REFERENCES

- [1] Eshtiak Ahmed, Ashrafur Islam, Atiqul Islam Chowdhury, Mohammad Masudur Rahman, Shahhaj Chowdhury, and Md Imran Hosen. 2021. Robo-friend: Can a social robot empathize with your feelings effectively?. In *Progress in Advanced Computing and Intelligent Engineering: Proceedings of ICACIE 2020*. Springer, Singapore, 777–788.
- [2] James Auger. 2014. Living with robots: A speculative design approach. *Journal of Human-Robot Interaction* 3, 1 (2014), 20–42.
- [3] Simon Baron-Cohen. 1995. The eye direction detector (EDD) and the shared attention mechanism (SAM): Two cases for evolutionary psychology. In *In C. Moore & P. J. Dunham (Eds.) Joint attention: Its origins and role in development*. Lawrence Erlbaum Associates, Inc., Mahwah, New Jersey, 41–59.
- [4] Andrew P Bayliss, Alexandra Frischen, Mark J Fenske, and Steven P Tipper. 2007. Affective evaluations of objects are influenced by observed gaze direction and emotional expression. *Cognition* 104, 3 (2007), 644–653.
- [5] Virginia Braun and Victoria Clarke. 2012. Thematic analysis. In *APA Handbook of Research Methods in Psychology: Vol. 2. Research Designs*, Harris Cooper (Ed.). American Psychological Association, Washington, 57–71.
- [6] C. Breazeal and B. Scassellati. 1999. How to build robots that make friends and influence people. In *Proceedings 1999 IEEE/RSJ International Conference on Intelligent Robots and Systems. Human and Environment Friendly Robots with High Intelligence and Emotional Quotients (Cat. No. 99CH36289)*, Vol. 2. IEEE, Piscataway, New Jersey, United States, 858–863 vol.2. <https://doi.org/10.1109/IROS.1999.812787>
- [7] F. Bu, A. Steptoe, and D. Fancourt. 2020. Who is lonely in lockdown? Cross-cohort analyses of predictors of loneliness before and during the COVID-19 pandemic. *Public Health* 186 (Sep 2020), 31–34. <https://doi.org/10.1016/j.puhe.2020.06.036>
- [8] Joseph Crawford, Kerry Butler-Henderson, Jürgen Rudolph, Bashar Malkawi, Matt Glowatz, Rob Burton, Paulo Magni, and Sophia Lam. 2020. COVID-19: 20 countries' higher education intra-period digital pedagogy responses. *Journal of Applied Learning & Teaching* 3, 1 (2020), 1–20.
- [9] K. Dautenhahn, S. Woods, C. Kaouri, M. L. Walters, Kheng Lee Koay, and I. Werry. 2005. What is a robot companion - friend, assistant or butler?. In *2005 IEEE/RSJ International Conference on Intelligent Robots and Systems. IEEE, Piscataway, New Jersey, United States, 1192–1197*. <https://doi.org/10.1109/IROS.2005.1545189>
- [10] Anna Dolot. 2020. The influence of COVID-19 pandemic on the remote work-an employee perspective.
- [11] Timon Elmer, Kieran Mephram, and Christoph Stadtfeld. 2020. Students under lockdown: Comparisons of students' social networks and mental health before and during the COVID-19 crisis in Switzerland. *Plos one* 15, 7 (2020), e0236337.
- [12] B.J. Fogg. 2003. Chapter 1 - Overview of captology. In *Persuasive Technology*, B.J. Fogg (Ed.). Morgan Kaufmann, San Francisco, 15 – 22. <https://doi.org/10.1016/B978-155860643-2/50003-2>
- [13] B.J. Fogg. 2003. Chapter 5 - Computers as persuasive social actors. In *Persuasive Technology*, B.J. Fogg (Ed.). Morgan Kaufmann, San Francisco, 89 – 120. <https://doi.org/10.1016/B978-155860643-2/50007-X>
- [14] Alexandra Frischen and Steven P Tipper. 2004. Orienting attention via observed gaze shift evokes longer term inhibitory effects: implications for social interactions, attention, and memory. *Journal of Experimental Psychology: General* 133, 4 (2004), 516.
- [15] E Scott Geller. 2016. The psychology of self-motivation. In *Applied Psychology*, E Scott Geller (Ed.). Cambridge University Press, New York, USA, 83–118.
- [16] Chloe A Hamza, Lexi Ewing, Nancy L Heath, and Abby L Goldstein. 2020. When social isolation is nothing new: A longitudinal study psychological distress during COVID-19 among university students with and without preexisting mental health concerns. *Canadian Psychology* 62, 1 (2020), 20–30.
- [17] Eriş Hüseyin and Sinem BARUT. 2020. The Effect of Feeling of Loneliness on Burnout Levels in University Students. *Kuramsal Eğitimbilim Dergisi* 13, 2 (2020), 369–383. <https://doi.org/10.30831/akukey.606671>
- [18] Kari Daniel Karjalainen, Anna Elisabeth Sofia Romell, Potchchara Ratsamee, Asim Evren Yantac, Morten Fjeld, and Mohammad Obaid. 2017. Social drone companion for the home environment: A user-centric exploration. In *Proceedings of the 5th International Conference on Human Agent Interaction*. Association for Computing Machinery, New York, NY, United States, 89–96.
- [19] Magdalena Klopotek. 2017. The Advantages and Disadvantages of Working Remotely from the Perspective of Young Employees. In *Management Challenges in a Network Economy: Proceedings of the MakeLearn and TIIM International Conference 2017*. ToKnowPress, Lublin, Poland, 535–535.
- [20] Stephen RH Langton and Vicki Bruce. 1999. Reflexive visual orienting in response to the social attention of others. *Visual cognition* 6, 5 (1999), 541–567.
- [21] Kwan Min Lee, Younbo Jung, Jaywoo Kim, and Sang Ryong Kim. 2006. Are physically embodied social agents better than disembodied social agents?: The effects of physical embodiment, tactile interaction, and people's loneliness in human-robot interaction. *International Journal of Human-Computer Studies* 64, 10 (2006), 962 – 973. <https://doi.org/10.1016/j.ijhcs.2006.05.002>
- [22] Minha Lee, Sander Ackermans, Nena van As, Hanwen Chang, Enzo Lucas, and Wijnand Jsselstein. 2019. Caring for Vincent: A Chatbot for Self-Compassion. In *Proceedings of the 2019 CHI Conference on Human Factors in Computing Systems (Glasgow, Scotland Uk) (CHI '19)*. Association for Computing Machinery, New York, NY, USA, 1–13. <https://doi.org/10.1145/3290605.3300932>

- [23] Nancy R Lockwood. 2003. *Work/life Balance: Challenges and Solutions*. Society for Human Resource Management, Alexandria, USA.
- [24] Hadis Malekie and Zeinab Farhoudi. 2015. Making Humanoid Robots More Acceptable Based on the Study of Robot Characters in Animation. *International Journal of Robotics and Automation* 4, 1 (2015), 63–72.
- [25] Chris Michaelides. 2009. Sonification of Robot Communication: A Case Study.
- [26] Chris Moore, Philip J Dunham, and Phil Dunham. 2014. *Joint attention: Its origins and role in development*. Psychology Press, New York, USA.
- [27] Omar Mubin, Catherine J Stevens, Suleman Shahid, Abdullah Al Mahmud, and Jian-Jie Dong. 2013. A review of the applicability of robots in education. *Journal of Technology in Education and Learning* 1, 209-0015 (2013), 13.
- [28] Aoife O'Donovan and Brian Hughes. 01 Oct. 2007. Social support and loneliness in college students: Effects on pulse pressure reactivity to acute stress. *International Journal of Adolescent Medicine and Health* 19, 4 (01 Oct. 2007), 523 – 528. <https://doi.org/10.1515/IJAMH.2007.19.4.523>
- [29] Amit Kumar Pandey and Rodolphe Gelin. 2018. A mass-produced sociable humanoid robot: Pepper: The first machine of its kind. *IEEE Robotics & Automation Magazine* 25, 3 (2018), 40–48.
- [30] Lawrence Robinson, Kai Lundgren, and Robert Segal. 2022. The Health and Mood-Boosting Benefits of Pets. <https://www.helpguide.org/articles/mental-health/mood-boosting-power-of-dogs.htm>
- [31] Namin Shin and Sangah Kim. 2007. Learning about, from, and with Robots: Students' Perspectives. In *RO-MAN 2007-The 16th IEEE International Symposium on Robot and Human Interactive Communication*. IEEE, Piscataway, New Jersey, United States, 1040–1045.
- [32] Litao Sun, Yongming Tang, and Wei Zuo. 2020. Coronavirus pushes education online. *Nature Materials* 19, 6 (2020), 687–687.
- [33] Ann Verhetsel, Roselinde Kessels, Toon Zijlstra, and Marjolein Van Bavel. 2017. Housing preferences among students: collective housing versus individual accommodations? A stated preference study in Antwerp (Belgium). *Journal of Housing and the Built Environment* 32, 3 (2017), 449–470.